

EXERCISES

LEVEL I

(Problems based on Fundamentals)

ORDER OF MATRICES

- Find the number of all possible matrices of order 2×2 with each entry 0 or 1.
- Find the number of all possible matrices of order 3×3 with each entry either 1 or 2.

ADDITION OF MATRICES

- If $A = \begin{pmatrix} 2 & 4 \\ 3 & 5 \end{pmatrix}$, find the additive inverse of A .

- Find a matrix X , if $X + \begin{pmatrix} 2 & 5 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 2 & 7 \end{pmatrix}$.

- Find X and Y , if $X + Y = \begin{pmatrix} 2 & 5 \\ 3 & -2 \end{pmatrix}$.

$$\text{and } X - Y = \begin{pmatrix} 4 & 2 \\ 8 & 2 \end{pmatrix}.$$

- Find a matrix X such that

$$A + 2B + X = O,$$

$$\text{where } A = \begin{pmatrix} 2 & -1 \\ 3 & 5 \end{pmatrix} \text{ and } B = \begin{pmatrix} -1 & 1 \\ 0 & 2 \end{pmatrix}.$$

- Find x and y , if

$$\begin{pmatrix} |x| & 2 \\ 5 & |y-2| \end{pmatrix} = \begin{pmatrix} <3 & 2 \\ 5 & <4 \end{pmatrix}$$

- Find $\Sigma(x+y)$, if

$$\begin{pmatrix} x^3 & 3x+2 & 2 \\ 3 & y^3+7y^2-35 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}$$

- Find x, y, z and t satisfying the equations

$$2 \begin{pmatrix} x & y \\ z & t \end{pmatrix} + 3 \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix} = 4 \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix}$$

- Find the matrices X and Y , if

$$2X + 3Y = \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix} \text{ and } 3X + 2Y = \begin{pmatrix} -1 & 2 \\ 1 & -5 \end{pmatrix}$$

- If $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$,

and $f(x) = 1 + x + x^2 + \dots$ to ∞ , find $f(A)$.

MULTIPLICATION OF MATRICES

- Let $A = [1 \ 2]$ and $B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

Find AB and BA .

- Let $A = [a \ b \ c]$ and $B = \begin{bmatrix} a^2 \\ b^2 \\ c^2 \end{bmatrix}$.

Find AB and BA .

- Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 4 \\ 5 & 7 \end{pmatrix}$.

Find AB and BA .

15. If A be a 2×3 matrix and AB a 2×5 matrix, find the order of the matrix B .

16. If A be a $(A)_{2 \times 3}$ matrix and B a $(B)_{3 \times 3}$ matrix and C a $(C)_{3 \times 7}$ matrix, find $2015ABC$.

17. If $A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$,

prove that

$$A^2 + 2A + I_2 = \mathbf{O}$$

18. If $A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $B = (2 \ 3 \ 4)$, AB and BA .

19. If $A = \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}$, find A^2 , A^3 , and A^4 .

20. Find a 2×2 matrix X such that

$$\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix} \cdot X = \begin{pmatrix} 3 & 5 \\ 0 & 6 \end{pmatrix}$$

21. If $A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$,

prove that $A^2 - 4A - 5I_3 = \mathbf{O}$.

22. Find x , if $(1 \ x \ 1) \begin{pmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ x \end{pmatrix} = 0$.

23. If $A = \begin{pmatrix} a & 0 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$, the value of a for which $A^2 = B$ is

- (a) 1 (b) -1
(c) 4 (d) No real values

24. If $A = \begin{pmatrix} \alpha & 2 \\ 2 & \alpha \end{pmatrix}$ and $|A^3| = 125$, the value of α is

- (a) ± 1 (b) ± 2 (c) ± 3 (d) ± 5

25. If A and B be two square matrices of order 3×3 which satisfy $AB = A$ and $BA = B$, find $(A + B)^7$.

26. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are two matrices such

that $AB = BA$ and $c \neq 0$, find the value of $\frac{a-d}{3b-c}$.

27. If A be a square matrix of order 2 such that $A^2 = \mathbf{O}$, find $A(I + A)^{2009}$.

28. If A be a square matrix and I be an identity matrix of the same order such that $A^2 = I$, $(I - A)(I + A)$.

29. If A be a square matrix such that $A^2 = A$, find $(I + A)^3 - 7A$.

30. If $A = \begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix}$ and $f(x) = \sum_{n=0}^{16} x^n$, find $f(A)$.

31. If A be a square matrix such that $A^2 = A + I$, where I is the identity matrix of the same order, find A^5 .

32. If A be a square matrix such that $A^2 = A - I$, where I is the identity matrix of the same order, find A^n .

33. If $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $n \in \mathbb{N}$, find A^n .

34. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$.

If U_1 and U_2 are column matrices such that

$$AU_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } AU_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \text{ find } U_1 + U_2.$$

SYMMETRIC AND SKEW-SYMMETRIC MATRICES

35. Express the matrix $A = \begin{pmatrix} 2 & 3 \\ 5 & 6 \end{pmatrix}$ as a sum of a symmetric and a skew-symmetric matrices.

36. Express the matrix $B = \begin{pmatrix} 1 & 3 & 4 \\ 2 & 3 & 5 \\ 4 & 3 & 6 \end{pmatrix}$ as a sum of a symmetric and a skew-symmetric matrices.

DETERMINANT

37. If A be a skew-symmetric matrix of odd order, find $\det(A)$.

38. If A be a skew-symmetric matrix of even order, find $\det(A)$.

39. If $A = (a_{ij})$ be a square matrix of order n such that $a_{ij} = \begin{cases} 0 & i = j \\ 1 & i \neq j \end{cases}$, find $\det(A)$.

40. If $A = \begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$ be a square matrix of order 2, find $\det(A^{2010} + 2A^{2009})$.

41. If A be a square matrix such that $A^2 = A$, find $\det(A)$.

42. If A be a square matrix of order 3 such that $\det(A) = 8$, find $\det(3A)$.

43. If A be a square matrix of order n such that $|A| = 2$, find $|An|$ for $n \in \mathbb{I}^+$.

44. Find the maximum value of the determinant

$$\begin{vmatrix} \cos^2 \theta & \sin^2 \theta \\ \sin^2 \theta & \cos^2 \theta \end{vmatrix}$$

45. Let P and Q be two square matrices of order 3 such that $P^3 = Q^3$ and $P^2Q = Q^2P$ and $P \neq Q$, find $\det(P^2 + Q^2)$.

46. Expand the determinant

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

47. Evaluate:

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$$

48. Evaluate:

$$\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$$

49. Evaluate:

$$\begin{vmatrix} \sin \alpha & \cos \beta & \cos(\alpha + \theta) \\ \sin \beta & \cos \beta & \cos(\beta + \theta) \\ \sin \gamma & \cos \gamma & \cos(\gamma + \theta) \end{vmatrix}$$

50. Prove that the value of

$$\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(a+c) \\ 1 & ab & c(a+b) \end{vmatrix}$$

51. Prove that

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3$$

52. Prove that

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

53. Prove that

$$\begin{vmatrix} b^2+c^2 & a^2 & a^2 \\ b^2 & c^2+a^2 & b^2 \\ c^2 & c^2 & a^2+b^2 \end{vmatrix} = 4a^2b^2c^2$$

54. Prove that

$$\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3a \end{vmatrix} = a^3$$

55. Prove that

$$\begin{vmatrix} 1+a^2 & b^2 & 2ab & 2b \\ 2ab & 1 & a^2+b^2 & 2a \\ 2b & 2a & 1 & a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$$

56. Prove that

$$\begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & cb & c^2+1 \end{vmatrix} = (1+a^2+b^2+c^2)$$

57. Prove that

$$\begin{vmatrix} {}^xC_1 & {}^xC_2 & {}^xC_3 \\ {}^yC_1 & {}^yC_2 & {}^yC_3 \\ {}^zC_1 & {}^zC_2 & {}^zC_3 \end{vmatrix} = \frac{xyz}{12} (x-y)(y-z)(z-x)$$

58. Prove that

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$$

$$= abc \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = abc + bc + ca + ab$$

59. Prove that

$$\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

60. Prove that

$$\begin{vmatrix} (a+1)(a+2) & (a+2) & 1 \\ (a+2)(a+3) & (a+3) & 1 \\ (a+3)(a+4) & (a+4) & 1 \end{vmatrix} = 2$$

61. Prove that

$$\begin{vmatrix} b+c & c+a & a+b \\ a+b & b+c & c+a \\ c+a & a+b & b+c \end{vmatrix} = 2 \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

62. Solve the following system of equations:

$$2x + 3y = 4$$

$$3x - 2y = 5$$

63. Solve the following system of equations:

$$x + 3y = 4$$

$$2x + 6y = 10$$

64. Solve the following system of equations:

$$2x + 5y = 6$$

$$6x + 15y = 18$$

 65. Find the number of triplets of a, b and c for which the system of equations

$$ax - by = 2a - b$$

$$(c+1)x + cy = 10 - a + 3b$$

has infinitely many solutions.

 66. Solve for x, y, z :

$$x + y + z = 1$$

$$ax + by + cz = d$$

$$a^2x + b^2y + c^2z = d$$

 67. Solve for x, y, z :

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{1}{4}$$

$$\frac{2}{x} + \frac{1}{y} + \frac{3}{z} = \frac{9}{4}$$

$$\frac{1}{x} + \frac{2}{y} + \frac{4}{z} = 1$$

 68. Find the equation of the parabola $y = ax^2 + bx + c$, which passes through the points $(2, 4)$, $(-1, 1)$ and $(-2, 5)$.

 69. Find the value of k , for which the system of equations

$$2x + ky = 5$$

$$3x - 4y = 7$$

has a unique solution.

70. Find the value of λ , for which the system of equations

$$3x + 4y = 5$$

$$\lambda x + 8y = 10$$

give infinitely many solutions.

71. If the system of equations

$$x + 2y + 3z = 1$$

$$(p + 2)z = 3$$

$$(2p + 1)y + z = 2$$

is inconsistent, find the value of p .

72. If the system of equations

$$2x + y + 2z = 2$$

$$x + 2y + z = 4$$

$$x + 2y + \lambda z = 4$$

has no solutions, find λ .

HOMOGENEOUS EQUATIONS

73. Solve the system of equations:

$$2x + 3y = 0$$

$$4x + 6y = 0.$$

74. Find the number of values of t for which the system of equations

$$(a - t)x + by + cz = 0$$

$$bx + (c - t)y + az = 0$$

$$cx + ay + (b - t)z = 0$$

has non-trivial solution.

75. Find the value of λ , if the system of equations

$$6x + 5y + \lambda z = 0$$

$$3x + y + 4z = 0$$

$$x + 2y + 3z = 0$$

has a unique solution.

76. If the system of equations $x + ay = 0$, $y + az = 0$ and $z + ax = 0$ has infinite solutions, find a .

MULTIPLICATION OF DETERMINANTS

77. Prove that

$$\begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix}^2 = \begin{vmatrix} c^2 + a^2 & a^2 & c^2 \\ a^2 & a^2 + b^2 & b^2 \\ c^2 & b^2 & b^2 + c^2 \end{vmatrix}.$$

78. Prove that

$$\begin{vmatrix} 2 & \alpha + \beta + \gamma + \delta & \alpha\beta + \gamma\delta \\ \alpha + \beta + \gamma + \delta & 2(\alpha + \beta)(\gamma + \delta) & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) \\ \alpha\beta + \gamma\delta & \alpha\beta(\gamma + \delta) + \gamma\delta(\alpha + \beta) & 2\alpha\beta\gamma\delta \end{vmatrix} = 0$$

79. Prove that

$$\begin{vmatrix} 1 & \cos(\beta - \alpha) & \cos(\gamma - \alpha) \\ \cos(\alpha - \beta) & 1 & \cos(\gamma - \beta) \\ \cos(\alpha - \gamma) & \cos(\beta - \gamma) & 1 \end{vmatrix} = 0$$

80. If $\alpha, \beta \neq 0$ and $f(n) = \alpha^n + \beta^n$ and

$$\begin{vmatrix} 3 & 1 + f(1) & 1 + f(2) \\ 1 + f(1) & 1 + f(2) & 1 + f(3) \\ 1 + f(2) & 1 + f(3) & 1 + f(4) \end{vmatrix} = k(1 - \alpha)^2(1 - \beta)^2(\alpha - \beta)^2$$

find k .

DIFFERENTIATION OF DETERMINANT

$$81. \text{ If } F(x) = \begin{vmatrix} 1 & a & a^2 \\ x & x^2 & x^3 \\ e^{x-a} & e^{x^2-a^2} & e^{x^3-a^3} \end{vmatrix},$$

find the value of $F'(a)$.

$$82. \text{ Let } f(x) = \begin{vmatrix} 3 & 2 & 1 \\ 6x^2 & 2x^3 & x^4 \\ 1 & b & b^2 \end{vmatrix},$$

find $f''(b)$.

INTEGRATION OF DETERMINANTS

$$83. \text{ If } f(x) = \begin{vmatrix} \sin^2 x & \log(\sin x) & \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \\ n & \sum_{k=1}^n (k) & \prod_{k=1}^n (k) \\ \frac{8}{15} & \frac{\pi}{2} \log\left(\frac{1}{2}\right) & \frac{\pi}{4} \end{vmatrix},$$

find the value of $\int_0^{\pi/2} f(x) dx$.

$$84. \text{ If } \Delta_r = \begin{vmatrix} r & 2012 & \frac{n(n+1)}{2} \\ 2r & 1 & 2013 \\ 3r & 2 & 2014 \end{vmatrix} \frac{n(3n-1)}{2}$$

find $\sum_{r=1}^n \Delta_r$

$$85. \text{ If } D_r = \begin{vmatrix} 2^{r-1} & 101 & (2^n - 1) \\ 3^{r-1} & 102 & \left(\frac{3^n - 1}{2}\right) \\ 5^{r-1} & 103 & \left(\frac{5^n - 1}{4}\right) \end{vmatrix},$$

find the value of $\sum_{r=1}^n D_r$

ADJOINT OF MATRICES

86. If A be a square matrix of order n , find $\text{adj}(A') - \text{adj } A'$.

87. If A be a non-singular matrix of order 3, find $\det(\text{adj } A^3)$.

88. If A be a square matrix of order 2 such that $A \cdot (\text{adj } A) = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}$, find $\det(A)$.
89. If A be a square matrix of order 3 such that $\det(A) = 4$, find $\det(\text{adj } A)$.
90. If A be a square matrix of order n such that $|\text{adj}(\text{adj } A)| = |A|^{16}$, find n .
91. If the adjoint of a matrix P of order 3 is $\begin{pmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{pmatrix}$, the possible values of the determinant of P is (are)
(a) -2 (b) -1 (c) 1 (d) 2
92. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j} a_{ij}$ for $1 \leq i, j \leq 3$.
If the determinant of P is 2, the determinant of the matrix Q is
(a) 2^{10} (b) 2^{11} (c) 2^{12} (d) 2^{13}
93. If $P = \begin{pmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{pmatrix}$ is the adjoint of a matrix A of order 3×3 and $|A| = 4$, find α .
94. Find the inverse of a skew symmetric matrix of odd order.
95. If B be a non-singular matrix and A is a square matrix of the same order, find $|B^{-1}AB|$.
96. If $P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $Q = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, find the inverse of a matrix R , such that $R = (P \cos \theta + Q \sin \theta)$.
97. If A and B be two non-singular square matrices such that $B^{-1}AB = A^2$, find $B^{-3}AB^3$.
98. If A be a non-singular matrix satisfying $I + A + A^2 + A^3 + \dots + A^k = O$, find A^{-1} .
99. If A be a non-singular matrix satisfying $A^2 - A + I = O$, find A^{-1} .
100. If A be a non-singular square matrix of order 3×3 such that $\det(A) = 5$, find $\det(\text{adj}(A^{-1}))$.
101. If A be a non-singular square matrix of order 3×3 , find $|A^{-1} \text{adj}(A)|$.
102. If $A = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$, find the multiplicative inverse of A .
103. If X be a non-singular square matrix of order 2 such that $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$, find X .
104. Let $A = \begin{pmatrix} 3 & 2 \\ 2 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 1 \\ 7 & 3 \end{pmatrix}$, find the value of $\det(2A^9B^{-1})$.

105. Let P and Q be two square matrices such that $|P| = 1 = |Q|$. If A and B be two square matrices of the same order such that $(\text{adj } B) = A$, find $\text{adj}(QBP)$.
106. If A is an 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1}A'$, find BB' . **[JEE Main, 2014]**

107. Find the inverse of the matrix, $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 4 \end{pmatrix}$.

108. If $A = \begin{pmatrix} 0 & -\tan\left(\frac{\alpha}{2}\right) \\ \tan\left(\frac{\alpha}{2}\right) & 0 \end{pmatrix}$,

prove that $(I + A)(I - A) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$.

109. Solve the following system of equations by matrix method.

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x + y - z = 1$$

110. Compute A^{-1} for the matrix

$$A = \begin{pmatrix} -1 & 2 & 5 \\ 2 & -3 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

and hence solve the following system of equations

$$x + 2y + 5z = 2$$

$$2x + 3y + z = 15$$

$$x + y + z = 3$$

111. Solve the following system of equations by matrix (inverse) method.

$$5x + 3y + 7z = 4$$

$$3x + 26y + 2z = 9$$

$$7x + 2y + 10z = 5$$

112. The number of 3×3 matrices A whose entries are either 0 or 1 and for which the system

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

has exactly two solutions, is

- (a) 0 (b) $2^9 - 1$ (c) 168 (d) 12

113. Let a, b, c be positive real numbers. Prove that the system of equations

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1, \frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

and $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

in x, y, z have a unique solutions.

114. By using elementary row operation, find the inverse of

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

RANK OF A MATRIX

115. Find the rank of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix}$.

116. Find the rank of $A = \begin{pmatrix} 2 & 4 & 3 \\ 1 & 2 & -1 \\ -1 & 2 & 6 \end{pmatrix}$.

117. Find the rank of $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{pmatrix}$.

118. Find the rank of $A = \begin{pmatrix} 1 & -2 & 1 & 1 \\ 1 & 1 & 2 & 3 \\ 4 & 1 & -5 & 6 \end{pmatrix}$.

ADVANCE MATRICES

119. Prove that the matrix $A = \begin{bmatrix} 2 & 3 & 5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix}$

is idempotent

120. Prove that the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -1 \end{bmatrix}$ is periodic.

121. Show that $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ 2 & 1 & 3 \end{bmatrix}$ is nilpotent matrix of order 3.

122. Show that the matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$

is involuntary.

123. Prove that the matrix $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$

is orthogonal matrix.

124. Prove that $A = \begin{bmatrix} 3 & 3 & 4i & 5+2i \\ 3+4i & 5 & 2+i \\ 5 & 2i & 2 & i \\ 2 & i & 2 & 2 \end{bmatrix}$

is an Hermitian matrix.

125. Prove that $A = \begin{bmatrix} 2i & 2 & 3i & 2+i \\ 2 & 3i & i & 3i \\ 2+i & 3i & 0 \end{bmatrix}$

is an skew-hermitian matrix.

126. Prove that the matrix $\frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 & 1+i \\ 1 & i & 1 \end{bmatrix}$ is unitary.

Hints & Solutions

4. We have,

$$X + \begin{pmatrix} 2 & 5 \\ 3 & -2 \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 2 & 7 \end{pmatrix}$$

$$\Rightarrow X = \begin{pmatrix} 3 & 6 \\ 2 & 7 \end{pmatrix} - \begin{pmatrix} 2 & 5 \\ 3 & -2 \end{pmatrix}$$

$$\Rightarrow X = \begin{pmatrix} 1 & 1 \\ -1 & 9 \end{pmatrix}$$

5. Clearly, $X = \frac{1}{2} \left(\begin{pmatrix} 2 & 5 \\ 3 & -2 \end{pmatrix} + \begin{pmatrix} 4 & 2 \\ 8 & -2 \end{pmatrix} \right)$

$$\Rightarrow X = \frac{1}{2} \begin{pmatrix} 6 & 7 \\ 11 & -4 \end{pmatrix} = \begin{pmatrix} 3 & 7/2 \\ 11/2 & -2 \end{pmatrix}$$

and $Y = \frac{1}{2} \left(\begin{pmatrix} 2 & 5 \\ 3 & -2 \end{pmatrix} - \begin{pmatrix} 4 & 2 \\ 8 & -2 \end{pmatrix} \right)$

$$\Rightarrow Y = \frac{1}{2} \begin{pmatrix} -2 & 3 \\ -5 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 3/2 \\ -5/2 & 0 \end{pmatrix}$$

6. Given $A + 2B + X = O$

$$\Rightarrow X = -(A + 2B)$$

$$= - \left(\begin{pmatrix} 2 & -1 \\ 3 & 5 \end{pmatrix} + 2 \begin{pmatrix} -1 & 1 \\ 0 & 2 \end{pmatrix} \right)$$

$$= - \left(\begin{pmatrix} 2 & -1 \\ 3 & 5 \end{pmatrix} + \begin{pmatrix} -2 & 2 \\ 0 & 4 \end{pmatrix} \right)$$

$$= - \begin{pmatrix} 0 & 1 \\ 3 & 9 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -3 & -9 \end{pmatrix}$$

7. Given $\begin{pmatrix} |x| & 2 \\ 5 & |y-2| \end{pmatrix} = \begin{pmatrix} <3 & 2 \\ 5 & <4 \end{pmatrix}$

$$\Rightarrow |x| < 2, |y-2| < 3$$

$$\Rightarrow 2 < x < 2, 3 < (y-2) < 3$$

$$\Rightarrow -2 < x < 2, -1 < y < 5$$

8. We have

$$\begin{pmatrix} x^3 - 3x + 2 & 2 \\ 3 & y^3 + 7y^2 - 35 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 3 & 1 \end{pmatrix}$$

$$x^3 - 3x + 2 = 0, \quad y^3 + 7y^2 - 35 = 1$$

$$x^3 - 3x + 2 = 0, \quad y^3 + 7y^2 - 36 = 1$$

$$\text{Now, } x^3 - 3x + 2 = 0$$

$$\Rightarrow x^3 - x^2 + x^2 - x - 2x + 2 = 1$$

$$\Rightarrow x^2(x-1) + x(x-1) - 2(x-1) = 0$$

$$\Rightarrow (x-1)(x^2 + x - 2) = 0$$

$$\Rightarrow (x-1)(x+2)(x-1) = 0$$

$$\Rightarrow (x-1)^2(x+2) = 0$$

$$\Rightarrow x = 1, 2$$

$$\text{Also, } y^3 + 7y^2 - 36 = 0$$

$$\Rightarrow y^3 - 2y^2 + 9y^2 - 18y + 18y - 36 = 1$$

$$\Rightarrow y^2(y-2) + 9y(y-2) + 18(y-2) = 0$$

$$\Rightarrow (y-2)(y^2 + 9y + 18) = 0$$

$$\Rightarrow (y-2)(y+3)(y+6) = 0$$

$$\Rightarrow y = 2, 3, 6$$

Thus,

$$\Sigma(x+y) = (1+2+2+3+6) = 8$$

9. We have

$$2 \begin{pmatrix} x & y \\ z & t \end{pmatrix} + 3 \begin{pmatrix} 1 & -2 \\ 0 & 4 \end{pmatrix} = 4 \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix}$$

$$\Rightarrow 2 \begin{pmatrix} x & y \\ z & t \end{pmatrix} = 4 \begin{pmatrix} 3 & 5 \\ 4 & 6 \end{pmatrix} - 3 \begin{pmatrix} 1 & -2 \\ 0 & 4 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2x & 2y \\ 2z & 2t \end{pmatrix} = \begin{pmatrix} 12 & 20 \\ 16 & 24 \end{pmatrix} - \begin{pmatrix} 3 & -6 \\ 0 & 12 \end{pmatrix}$$

$$= \begin{pmatrix} 9 & 26 \\ 16 & 12 \end{pmatrix}$$

$$\Rightarrow x = 9/2, y = 13, z = 8, t = 6$$

10. Let $A = \begin{pmatrix} 2 & 3 \\ 4 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ 1 & -5 \end{pmatrix}$

Solving, we get

$$X = \frac{1}{5}(3B - 2A) \text{ and } Y = \frac{1}{5}(3A - 2B)$$

$$\text{Thus, } X = \frac{1}{5} \left(\begin{pmatrix} -3 & 6 \\ 3 & -15 \end{pmatrix} - \begin{pmatrix} 4 & 6 \\ 8 & 0 \end{pmatrix} \right)$$

$$\Rightarrow X = \frac{1}{5} \begin{pmatrix} -7 & 0 \\ -5 & -15 \end{pmatrix} = \begin{pmatrix} -7/5 & 0 \\ -1 & -3 \end{pmatrix}$$

and $Y = \frac{1}{5} \left(\begin{pmatrix} 6 & 9 \\ 12 & 0 \end{pmatrix} - \begin{pmatrix} -2 & 4 \\ 2 & -10 \end{pmatrix} \right)$

$$\Rightarrow Y = \frac{1}{5} \begin{pmatrix} 8 & 5 \\ 10 & 10 \end{pmatrix} = \begin{pmatrix} 8/5 & 1 \\ 2 & 2 \end{pmatrix}$$

11. Given $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$

$$= 2 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 2I$$

We have,

$$f(x) = 1 + x + x^2 + \dots \text{ to } \infty$$

$$= \frac{1}{1-x}$$

$$\Rightarrow f(A) = \frac{I}{I-A}$$

$$= \frac{I}{I-2I}$$

$$= -\frac{I}{I}$$

$$= -\frac{I^2}{I}$$

$$= -I$$

12. Then $AB = [1 \times 2 + 2 \times 3] = [8]$

13. Then $AB = [a \times a^2 + b \times b^2 + c \times c^2]$
 $= [a^3 + b^3 + c^3]$

14. Then $AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \times \begin{pmatrix} 2 & 4 \\ 5 & 7 \end{pmatrix}$
 $= \begin{pmatrix} 1.2 + 2.5 & 1.4 + 2.7 \\ 3.2 + 4.5 & 3.4 + 4.7 \end{pmatrix}$
 $= \begin{pmatrix} 12 & 18 \\ 26 & 40 \end{pmatrix}$

Also,

$$\begin{pmatrix} 2 & 4 \\ 5 & 7 \end{pmatrix} \times \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 2.1 + 4.3 & 2.2 + 4.7 \\ 5.1 + 7.3 & 5.2 + 7.4 \end{pmatrix}$$

$$= \begin{pmatrix} 14 & 32 \\ 26 & 38 \end{pmatrix}$$

Thus, $AB \neq BA$

Clearly, the matrix multiplication is not commutative.

15. Given A is a 2×3 matrix and AB is a 2×5 matrix.

Thus, B is a matrix of 3×5 .

16. Clearly, the matrix multiplication is not defined.

17. Given $A = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$

Now, $A^2 = A.A$

$$= \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

Also

$$A^2 - 2A + I_2 = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} - 2 \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} + \begin{pmatrix} -2 & 0 \\ 2 & -2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$= 0$$

Hence, the result

18. We have,

$$AB = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} (2 \ 3 \ 4)$$

$$= \begin{pmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{pmatrix}$$

Also,

$$BA = (2 \ 3 \ 4) \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$= (2 + 6 + 12) = (20)$$

19. We have,

$$A^2 = A.A = \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4+12 & 6+3 \\ 8+4 & 12+1 \end{pmatrix} = \begin{pmatrix} 16 & 9 \\ 12 & 13 \end{pmatrix}$$

Now, $A^3 = A^2.A$

$$= \begin{pmatrix} 16 & 9 \\ 12 & 13 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 32+36 & 48+9 \\ 24+52 & 36+39 \end{pmatrix} = \begin{pmatrix} 68 & 57 \\ 76 & 69 \end{pmatrix}$$

Also, $A^4 = A^3.A$

$$= \begin{pmatrix} 68 & 57 \\ 76 & 69 \end{pmatrix} \begin{pmatrix} 2 & 3 \\ 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 136+228 & 204+57 \\ 152+276 & 228+69 \end{pmatrix} = \begin{pmatrix} 364 & 261 \\ 428 & 297 \end{pmatrix}$$

20. Let $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

So, $\begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 0 & 6 \end{pmatrix}$

$$\Rightarrow \begin{pmatrix} 2a+c & 2b+d \\ a+4c & b+4d \end{pmatrix} = \begin{pmatrix} 3 & 5 \\ 0 & 6 \end{pmatrix}$$

$$2a+c=3 \quad 2b+d=5$$

$$\Rightarrow a+4c=0 \quad b+4d=6$$

$$\Rightarrow -8c+c=3, 2(6-4d)+d=5$$

$$\Rightarrow c = -\frac{3}{7}, -7d = 5 - 12 = -7$$

$$\Rightarrow c = -\frac{3}{7}, d = 1$$

Therefore, $a = \frac{12}{7}, b = 2$

Thus, the matrix $X = \begin{pmatrix} 12/7 & 2 \\ -3/7 & 1 \end{pmatrix}$.

21. We have,

$$A^2 = A.A = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix}$$

$$\begin{aligned}
 &= \begin{pmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{pmatrix} \\
 &= \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix}
 \end{aligned}$$

Now, $A^2 - 4A - 5I_3$

$$\begin{aligned}
 &= \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - 4 \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - 5 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - \begin{pmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} \\
 &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0
 \end{aligned}$$

22. We have

$$\begin{aligned}
 &(1 \ x \ 1) \begin{pmatrix} 1 & 3 & 2 \\ 2 & 5 & 1 \\ 15 & 3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ x \end{pmatrix} = 0 \\
 \Rightarrow &(1 \ x \ 1) \begin{pmatrix} 1+6+2x \\ 2+10+x \\ 15+6+2x \end{pmatrix} = 0 \\
 \Rightarrow &(1 \ x \ 1) \begin{pmatrix} 7+2x \\ 12+x \\ 21+2x \end{pmatrix} = 0 \\
 \Rightarrow &(7+2x) + x(12+x) + (21+x) = 0 \\
 \Rightarrow &(7+2x) + 12x + x^2 + (21+x) = 0 \\
 \Rightarrow &x^2 + 15x + 28 = 0 \\
 \Rightarrow &x = \frac{-15 \pm \sqrt{225 - 112}}{2} = \frac{-15 \pm \sqrt{113}}{2}
 \end{aligned}$$

Hence, the solution set is

$$\left\{ \frac{-15 + \sqrt{113}}{2}, \frac{-15 + \sqrt{113}}{2} \right\}$$

23. We have

$$A^2 = A \cdot A = \begin{pmatrix} a & 0 \\ 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} a & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} a^2 & 0 \\ a+1 & 1 \end{pmatrix}$$

Given relation is

$$A^2 = B$$

$$\Rightarrow \begin{pmatrix} a^2 & 0 \\ a+1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$$

$$\Rightarrow a^2 = 1, a+1 = 5$$

$$\Rightarrow a = \pm 1, a = 4$$

24. We have $A^2 = A \cdot A$

$$\begin{aligned}
 &= \begin{pmatrix} \alpha & 2 \\ 2 & \alpha \end{pmatrix} \cdot \begin{pmatrix} \alpha & 2 \\ 2 & \alpha \end{pmatrix} \\
 &= \begin{pmatrix} \alpha^2 + 4 & 4\alpha \\ 4\alpha & \alpha^2 + 4 \end{pmatrix}
 \end{aligned}$$

Now, $A^3 = A^2 \cdot A$

$$\begin{aligned}
 &= \begin{pmatrix} \alpha^2 + 4 & 4\alpha \\ 4\alpha & \alpha^2 + 4 \end{pmatrix} \cdot \begin{pmatrix} \alpha & 2 \\ 2 & \alpha \end{pmatrix} \\
 &= \begin{pmatrix} \alpha^3 + 12\alpha & 6\alpha^2 + 8 \\ 6\alpha^2 + 8 & \alpha^3 + 12\alpha \end{pmatrix}
 \end{aligned}$$

Given $|A^3| = 125$

$$\Rightarrow \begin{vmatrix} \alpha^3 + 12\alpha & 6\alpha^2 + 8 \\ 6\alpha^2 + 8 & \alpha^3 + 12\alpha \end{vmatrix} = 125$$

$$\Rightarrow (\alpha^3 + 12\alpha)^2 - (6\alpha^2 + 8)^2 = 125$$

$$\Rightarrow (\alpha^3 + 12\alpha + 6\alpha^2 + 8)(\alpha^3 + 12\alpha - 6\alpha^2 - 8) = 125$$

$$\Rightarrow (\alpha^3 + 6\alpha^2 + 12\alpha + 8)(\alpha^3 + 6\alpha^2 + 12\alpha - 8) = 125$$

$$\Rightarrow (\alpha + 2)^3(\alpha - 2)^3 = 125$$

$$\Rightarrow \{(\alpha + 2)(\alpha - 2)\}^3 = (5)^3$$

$$\Rightarrow (\alpha + 2)(\alpha - 2) = 5$$

$$\Rightarrow \alpha^2 - 4 = 5$$

$$\Rightarrow \alpha^2 = 9$$

$$\Rightarrow \alpha = \pm 3$$

25. Given $AB = A$ and $BA = B$

$$\text{Now } A = AB$$

$$\Rightarrow A^2 = ABA = AB = A$$

Similarly $B^2 = B$

We have,

$$\begin{aligned}
 (A+B)^2 &= (A^2 + AB + BA + B^2) \\
 &= (A + A + B + B) \\
 &= 2(A+B)
 \end{aligned}$$

$$\Rightarrow (A+B)^4 = 4(A+B)^2 = 8(A+B)$$

$$\text{and } (A+B)^3 = (A+B)^2 \cdot (A+B)$$

$$= 2(A+B) \cdot (A+B) = 2(A+B)^2$$

$$= 4(A+B)$$

Thus,

$$\begin{aligned}
 (A+B)^7 &= (A+B)^4 \cdot (A+B)^3 \\
 &= 32(A+B)^2 = 64(A+B)
 \end{aligned}$$

$$26. \text{ Given } A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \text{ and } B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\begin{aligned}
 \text{Now, } AB &= \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \\
 &= \begin{pmatrix} a+2c & b+2d \\ 3a+4c & 3b+4d \end{pmatrix}
 \end{aligned}$$

7.36

$$\begin{aligned}\text{Also, } BA &= \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} a+3b & 2a+4b \\ c+3d & 2c+4d \end{pmatrix}\end{aligned}$$

It is given that

$$AB = BA$$

$$\Rightarrow \begin{pmatrix} a+2c & b+2d \\ 3a+4c & 3b+4d \end{pmatrix} = \begin{pmatrix} a+3b & 2a+4b \\ c+3d & 2c+4d \end{pmatrix}$$

$$a+2c = a+3b \Rightarrow 2c = 3b$$

$$b+2d = 2a+4b$$

$$3a+4c = c+3d$$

$$\Rightarrow 3b+4d = 2c+4d$$

$$2c = 3b$$

$$\Rightarrow 2a-2d = -3b$$

$$\text{Now, } \frac{a-d}{3b-c} = \frac{-\frac{3}{2}b}{2c-c} = -\frac{\frac{3}{2}b}{c} = -\frac{\frac{3}{2}b}{\frac{3}{2}b} = -1$$

27. Given $A^2 = \mathbf{O}$

$$A^2 = \mathbf{O} = A^3 = A^4 = \dots = A^{2009}$$

$$\text{Now, } A(I+A)^{2009}$$

$$= A \left({}^{2009}C_0 \cdot I^n + {}^{2009}C_1 \cdot I^{n-1} \cdot A + {}^{2009}C_2 \cdot I^{n-2} \cdot A^2 + \dots + {}^{2009}C_{2009} \cdot I^{n-2009} \cdot A^{2008} \right)$$

$$= A(I + 2009A + \mathbf{O} + \mathbf{O} + \dots + \mathbf{O})$$

$$= A(I + 2009A)$$

$$= A \cdot I + 2009A^2$$

$$= A + \mathbf{O}$$

$$= A$$

28. Given $A^2 = I$.

Now,

$$(I-A)(I+A) = I^2 + IA - AI - A^2$$

$$= I + A - A - A^2$$

$$= I + A - A - I$$

$$= \mathbf{O}$$

29. We have,

$$(I+A)^3 - 7A = (I^3 + 3I^2A + 3IA^2 + A^3) - 7A$$

$$= (I + 3A + 3A^2 + A^3) - 7A$$

$$= (I + 3A + 3A + A) - 7A$$

$$= I + 7A - 7A$$

$$= I$$

30. Here, $A^2 = A \cdot A$.

$$= \begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = \mathbf{O}$$

$$\text{Thus, } A^2 = \mathbf{O} = A^3 = A^4 = \dots = A^{16}.$$

$$\text{We have } f(x) = \sum_{n=0}^{16} x^n$$

$$= 1 + x + x^2 + x^3 + \dots + x^{16}$$

$$\text{Now, } f(A) = I + A + A^2 + A^3 + \dots + A^{16}$$

$$= I + A + \mathbf{O} + \mathbf{O} + \dots + \mathbf{O}$$

$$= I + A$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 5 \\ 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 5 \\ 0 & 1 \end{pmatrix}$$

31. Given,

$$A^2 = A + I$$

$$\Rightarrow A^3 = A^2 + A = A + I + A = 2A + I$$

$$\Rightarrow A^4 = 2A^2 + A = 2(A + I) + A = 3A + 2I$$

$$\Rightarrow A^5 = 3A^2 + 2A = 3(A + I) + 2A = 5A + 3I$$

32. Given,

$$A^2 = 2A - I$$

$$\Rightarrow A^3 = 2A^2 - A = 2(2A - I) - A = 3A - 2I$$

$$\Rightarrow A^4 = 3A^2 - 2A = 3(2A - I) - 2A = 4A - 3I$$

$$\Rightarrow A^5 = 4A^2 - 3A = 4(2A - I) - 3A = 5A - 4I$$

Thus, by symmetry, we can say that

$$A^n = nA - (n-1)I$$

33. Given,

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\Rightarrow A^2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2 \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 2A$$

$$\Rightarrow A^3 = 2A^2 = 2(2A) = 4A = 2^2A$$

$$\Rightarrow A^4 = 2^2A^2 = 2^2(2A) = 2^3A$$

Thus, by symmetry, we can say that

$$A^n = 2^{n-1}A$$

$$34. \text{ Let } U_1 = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \text{ and } U_2 = \begin{pmatrix} d \\ e \\ f \end{pmatrix}$$

$$\text{Given } AU_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a \\ 2a+b \\ 3a+2b+c \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow U_1 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

Similarly, we can easily find that

$$U_2 = \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$$

$$\text{Thus, } U_1 + U_2 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

$$37. \text{ Let } A = \begin{pmatrix} 0 & 2 & 3 \\ -2 & 0 & 4 \\ 3 & 4 & 0 \end{pmatrix}$$

$$\begin{aligned} \text{Then } |A| &= \begin{vmatrix} 0 & 2 & 3 \\ 2 & 0 & 4 \\ 3 & 4 & 0 \end{vmatrix} \\ &= \begin{vmatrix} 0 & 4 \\ -4 & 0 \end{vmatrix} - 2 \begin{vmatrix} -2 & 4 \\ -3 & 0 \end{vmatrix} + 3 \begin{vmatrix} -2 & 0 \\ -3 & -4 \end{vmatrix} \\ &= 0 - 2(0 + 12) + 3(8 - 0) \\ &= -24 + 24 = 0 \end{aligned}$$

Thus, the determinant of skew-symmetric matrix of odd order is zero.

$$38. \text{ Let } A = \begin{pmatrix} 0 & b \\ b & 0 \end{pmatrix}$$

$$\text{Then } |A| = \begin{vmatrix} 0 & b \\ b & 0 \end{vmatrix} = b^2$$

Thus, the determinant of skew-symmetric matrix of even order is a perfect square.

$$39. \text{ Let } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\begin{aligned} \text{Then } |A| &= \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1 \\ &= (2 - 1)(-1)^{2-1} \end{aligned}$$

$$\begin{aligned} \text{Again, let } A &= \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \text{Thus, } |A| &= \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} \\ &= 0 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1 + 1 = 2 \\ &= (3 - 1)(-1)^{3-1} \end{aligned}$$

Therefore, in general

$$\begin{aligned} |A| &= \begin{vmatrix} 0 & 1 & \dots & 1 \\ 1 & 0 & \dots & 1 \\ \dots & \dots & \dots & \dots \\ 1 & 1 & \dots & 0 \end{vmatrix} \\ &= (n - 1)(-1)^{n-1} \end{aligned}$$

40. We have,

$$\begin{aligned} (A^{2016} + 2A^{2015}) &= A^{2015} (A + 2I_2) \\ &= A^{2015} \left[\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \right] \\ &= A^{2015} \begin{pmatrix} 4 & 5 \\ 1 & 5 \end{pmatrix} \end{aligned}$$

Thus,

$$\begin{aligned} |A^{2016} + 2A^{2015}| &= \left| A^{2015} \begin{pmatrix} 4 & 5 \\ 1 & 5 \end{pmatrix} \right| \\ &= |A|^{2015} \begin{vmatrix} 4 & 5 \\ 1 & 5 \end{vmatrix} \\ &= |A|^{2015} \times \begin{vmatrix} 4 & 5 \\ 1 & 5 \end{vmatrix} \\ &= (1)^{2015} \times (20 - 5) \\ &= 15 \end{aligned}$$

41. Given,

$$\begin{aligned} A &= A^2 \\ \Rightarrow |A| &= |A^2| \\ \Rightarrow |A| &= |A|^2 \\ \Rightarrow |A|(|A| - 1) &= 0 \\ \Rightarrow |A| &= 0 \text{ or } |A| = 1 \end{aligned}$$

42. We have,

$$\begin{aligned} \det(3A) &= 3^3 \times \det(A) \\ &= 3^3 \times 8 = 216 \end{aligned}$$

43. We have,

$$|A^n| = |A|^n = 2^n$$

44. We have,

$$\begin{aligned} \begin{vmatrix} \cos^2 \theta & \sin^2 \theta \\ \sin^2 \theta & \cos^2 \theta \end{vmatrix} &= \cos^4 \theta - \sin^2 \theta \\ &= (\cos^2 \theta - \sin^2 \theta)(\cos^2 \theta + \sin^2 \theta) \\ &= (\cos^2 \theta - \sin^2 \theta) \\ &= \cos(2\theta) \end{aligned}$$

Thus, the maximum value is 1.

45. Given $P^3 = Q^3$ and $P^2Q = Q^2P$

We have,

$$\begin{aligned} P^3 - P^2Q &= Q^3 - Q^2P \\ \Rightarrow P^2(P - Q) &= Q^2(Q - P) \\ \Rightarrow P^2(P - Q) &= -Q^2(P - Q) \\ \Rightarrow (P^2 + Q^2)(P - Q) &= \mathbf{O} \\ \Rightarrow (P^2 + Q^2) &= \mathbf{O}, \quad \text{since } (P - Q) \neq \mathbf{O} \end{aligned}$$

Thus, $\det(P^2 + Q^2) = \det(\mathbf{O}) = 0$

46. We have,

$$\begin{aligned} \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} &= a \begin{vmatrix} c & a \\ a & b \end{vmatrix} - b \begin{vmatrix} b & a \\ c & b \end{vmatrix} + c \begin{vmatrix} b & c \\ c & a \end{vmatrix} \\ &= a(bc - a^2) - b(b^2 - ac) + c(ab - c^2) \\ &= abc - a^3 - b^3 + abc + abc - c^3 \\ &= 3abc - a^3 - b^3 - c^3 \\ &= -(a^3 + b^3 + c^3 - 3abc) \end{aligned}$$

47. The given determinant = $\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$

$$\begin{vmatrix} 1 & a & a^2 \\ 0 & b-a & b^2-a^2 \\ 0 & c-a & c^2-a^2 \end{vmatrix} \quad \begin{matrix} (R_2 \rightarrow R_2 - R_1) \\ (R_3 \rightarrow R_3 - R_1) \end{matrix}$$

$$= \begin{vmatrix} b-a & b^2-a^2 \\ c-a & c^2-a^2 \end{vmatrix}$$

$$= (b-a)(c-a) \begin{vmatrix} 1 & b+a \\ 1 & c+a \end{vmatrix}$$

$$= (b-a)(c-a)(c+a-b-a)$$

$$= (b-a)(c-a)(c-b)$$

$$= (a-b)(b-c)(c-a)$$

48. The given determinant = $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$

$$\begin{vmatrix} 1 & a & a+b+c \\ 1 & b & b+c+a \\ 1 & c & c+a+b \end{vmatrix} \quad (C_3 \rightarrow C_2 + C_3)$$

$$= (a+b+c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix}$$

$$= (a+b+c) \times 0$$

$$= 0$$

49. The given determinant is

$$\begin{vmatrix} \sin \alpha & \cos \beta & \cos(\alpha + \theta) \\ \sin \beta & \cos \beta & \cos(\beta + \theta) \\ \sin \gamma & \cos \gamma & \cos(\gamma + \theta) \end{vmatrix}$$

$$\begin{vmatrix} \sin \alpha & \cos \beta & 0 \\ \sin \beta & \cos \beta & 0 \\ \sin \gamma & \cos \gamma & 0 \end{vmatrix}$$

$$[C_3 \rightarrow C_3 - (C_1 \cos \theta + C_2 \sin \theta)]$$

$$0$$

50. The given determinant = $\begin{vmatrix} 1 & bc & a(b+c) \\ 1 & ca & b(a+c) \\ 1 & ab & c(a+b) \end{vmatrix}$

$$\begin{vmatrix} 1 & bc & bc+ab+ac \\ 1 & ca & ac+ab+bc \\ 1 & ab & ab+ac+bc \end{vmatrix} \quad (C_3 \rightarrow C_2 + C_3)$$

$$= (ab+bc+ca) \times \begin{vmatrix} 1 & bc & 1 \\ 1 & ca & 1 \\ 1 & ab & 1 \end{vmatrix}$$

$$= (ab+bc+ca) \times 0$$

$$= 0$$

51. The given determinant

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix}$$

$$\begin{vmatrix} 2(a+b+c) & a & b \\ 2(a+b+c) & b+c+2a & b \\ 2(a+b+c) & a & c+a+2b \end{vmatrix}$$

$$(C_1 \rightarrow C_1 + C_2 + C_3)$$

$$= 2(a+b+c) \times \begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c) \times \begin{vmatrix} 1 & a & b \\ 0 & b+c+a & 0 \\ 0 & 0 & c+a+b \end{vmatrix}$$

$$\begin{matrix} (R_2 \rightarrow R_2 - R_1) \\ (R_3 \rightarrow R_3 - R_1) \end{matrix}$$

$$= 2(a+b+c)^3 \times \begin{vmatrix} 1 & a & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= 2(a+b+c)^3 \times 1$$

$$= 2(a+b+c)^3$$

52. The given determinant

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$\begin{vmatrix} 2(b+c) & 2(c+a) & 2(a+b) \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$(R_1 \rightarrow R_1 + R_2 + R_3)$$

$$2 \begin{vmatrix} (b+c) & (c+a) & (a+b) \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$= 2 \begin{vmatrix} b+c & (c+a) & (a+b) \\ -c & 0 & -a \\ -b & -a & 0 \end{vmatrix}$$

$$\begin{matrix} (R_2 \rightarrow R_2 - R_1) \\ (R_3 \rightarrow R_3 - R_1) \end{matrix}$$

$$= 2 \begin{vmatrix} 0 & c & b \\ -c & 0 & -a \\ -b & -a & 0 \end{vmatrix}$$

$$(R_1 \rightarrow R_1 + R_2 + R_3)$$

$$= 2[-c(-ab-0) + b(ca-0)]$$

$$= 2(abc + abc)$$

$$= 4abc$$

53. The given determinant

$$= \begin{vmatrix} b^2 + c^2 & a^2 & a^2 \\ b^2 & c^2 + a^2 & b^2 \\ c^2 & c^2 & a^2 + b^2 \end{vmatrix}$$

$$= \begin{vmatrix} 2(b^2 + c^2) & 2(c^2 + a^2) & 2(a^2 + b^2) \\ b^2 & c^2 + a^2 & b^2 \\ c^2 & c^2 & a^2 + b^2 \end{vmatrix}$$

$$(R_1 \rightarrow R_1 + R_2 + R_3)$$

$$= 2 \begin{vmatrix} (b^2 + c^2) & (c^2 + a^2) & (a^2 + b^2) \\ b^2 & c^2 + a^2 & b^2 \\ c^2 & c^2 & a^2 + b^2 \end{vmatrix}$$

$$= 2 \begin{vmatrix} b^2 + c^2 & (c^2 + a^2) & (a^2 + b^2) \\ c^2 & 0 & a^2 \\ b^2 & a^2 & 0 \end{vmatrix}$$

$$\begin{pmatrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{pmatrix}$$

$$= 2 \begin{vmatrix} 0 & c^2 & b^2 \\ -c^2 & 0 & a^2 \\ -b^2 & -a^2 & 0 \end{vmatrix} \quad (R_1 \rightarrow R_1 + R_2 + R_3)$$

$$= 2(c^2(a^2b^2 - 0) + b^2(c^2a^2 - 0))$$

$$= 2(a^2b^2c^2 + a^2b^2c^2)$$

$$= 4a^2b^2c^2$$

54. The given determinant

$$= \begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix}$$

$$= \begin{vmatrix} a & a+b & a+b+c \\ 0 & a & 2a+b \\ 0 & 3a & 7a+3b \end{vmatrix} \quad \begin{pmatrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{pmatrix}$$

$$= a \begin{vmatrix} a & 2a+b \\ 3a & 7a+3b \end{vmatrix}$$

$$= a(7a^2 + 3ab - 6a^2 - 3ab)$$

$$= a(a^2) = a^3$$

55. The given determinant

$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1+a^2-b^2+2b^2 & 2ab & 2ab & -2b \\ 2ab & 2ab & 1-a^2+b^2+2a^2 & 2a \\ 2b & b(1-a^2-b^2) & -2a+a(1-a^2-b^2) & 1-a^2-b^2 \end{vmatrix}$$

$$\begin{pmatrix} C_1 \rightarrow C_1 - bC_3 \\ C_2 \rightarrow C_2 + aC_3 \end{pmatrix}$$

$$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b(1+a^2+b^2) & -a(1+a^2+b^2) & 1-a^2-b^2 \end{vmatrix}$$

$$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & 2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$

$$= (1+a^2+b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ 0 & -a & 1-a^2+b^2 \end{vmatrix}$$

$$(R_3 \rightarrow R_3 - bR_1)$$

$$(1+a^2+b^2)^2 \begin{vmatrix} 1 & 2a \\ -a & 1-a^2+b^2 \end{vmatrix}$$

$$= (1+a^2+b^2)^2 (1-a^2+b^2+2a^2)$$

$$= (1+a^2+b^2)^3$$

56. The given determinant

$$= \begin{vmatrix} a^2+1 & ab & ac \\ ab & b^2+1 & bc \\ ac & cb & c^2+1 \end{vmatrix}$$

$$= \frac{1}{abc} \begin{vmatrix} a(a^2+1) & a^2b & a^2c \\ ab^2 & b(b^2+1) & b^2c \\ ac^2 & c^2b & c(c^2+1) \end{vmatrix}$$

$$= \frac{abc}{abc} \begin{vmatrix} (a^2+1) & a^2 & a^2 \\ b^2 & (b^2+1) & b^2 \\ c^2 & c^2 & (c^2+1) \end{vmatrix}$$

$$= \begin{vmatrix} (1+a^2+b^2+c^2) & (1+a^2+b^2+c^2) & (1+a^2+b^2+c^2) \\ b^2 & (b^2+1) & b^2 \\ c^2 & c^2 & (c^2+1) \end{vmatrix}$$

$$(R_1 \rightarrow R_1 + R_2 + R_3)$$

$$= (1+a^2+b^2+c^2) \begin{vmatrix} 1 & 1 & 1 \\ b^2 & (b^2+1) & b^2 \\ c^2 & c^2 & (c^2+1) \end{vmatrix}$$

$$= (1+a^2+b^2+c^2) \begin{vmatrix} 1 & 0 & 0 \\ b^2 & 1 & 0 \\ c^2 & 0 & 1 \end{vmatrix} \quad \begin{pmatrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{pmatrix}$$

$$= (1 + a^2 + b^2 + c^2) \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$= (1 + a^2 + b^2 + c^2)$$

57. The given determinant

$$\begin{vmatrix} {}^xC_1 & {}^xC_2 & {}^xC_3 \\ {}^yC_1 & {}^yC_2 & {}^yC_3 \\ {}^zC_1 & {}^zC_2 & {}^zC_3 \end{vmatrix}$$

$$= \begin{vmatrix} x & \frac{x(x-1)}{2} & \frac{x(x-1)(x-2)}{6} \\ y & \frac{y(y-1)}{2} & \frac{y(y-1)(y-2)}{6} \\ z & \frac{z(z-1)}{2} & \frac{z(z-1)(z-2)}{6} \end{vmatrix}$$

$$= \frac{xyz}{12} \begin{vmatrix} 1 & (x-1) & (x-1)(x-2) \\ 1 & (y-1) & (y-1)(y-2) \\ 1 & (z-1) & (z-1)(z-2) \end{vmatrix}$$

$$= \frac{xyz}{12} \begin{vmatrix} 1 & (x-1) & (x-1)(x-2) \\ 0 & (y-x) & (y^2-x^2)-3(y-x) \\ 0 & (z-x) & (z^2-x^2)-3(z-x) \end{vmatrix}$$

$$\left(\begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \right)$$

$$= \frac{xyz}{12} \begin{vmatrix} (y-x) & (y^2-x^2) & 3(y-x) \\ (z-x) & (z^2-x^2) & 3(z-x) \end{vmatrix}$$

$$= \frac{xyz(y-x)(z-x)}{12} \begin{vmatrix} 1 & y+x & 3 \\ 1 & z+x & 3 \end{vmatrix}$$

$$= \frac{xyz(y-x)(z-x)(z-y)}{12}$$

$$= \frac{xyz(x-y)(y-z)(z-x)}{12}$$

58. The given determinant

$$= \begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix}$$

$$= abc \begin{vmatrix} 1+\frac{1}{a} & \frac{1}{a} & \frac{1}{a} \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} \end{vmatrix}$$

$$= abc \begin{vmatrix} 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} & 1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} \end{vmatrix}$$

$$= abc \left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \right) \begin{vmatrix} 1 & 1 & 1 \\ \frac{1}{b} & 1+\frac{1}{b} & \frac{1}{b} \\ \frac{1}{c} & \frac{1}{c} & 1+\frac{1}{c} \end{vmatrix}$$

$$= abc \left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \right) \begin{vmatrix} 1 & 0 & 0 \\ \frac{1}{b} & 1 & 0 \\ \frac{1}{c} & 0 & 1 \end{vmatrix}$$

$$= abc \left(1+\frac{1}{a}+\frac{1}{b}+\frac{1}{c} \right)$$

59. The given determinant is

$$= \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix}$$

$$= \begin{vmatrix} (b+c)^2 & a^2 & 0 & a^2 \\ 0 & (c+a)^2 & b^2 & b^2 \\ c^2-(a+b)^2 & c^2-(a+b)^2 & (a+b)^2 \end{vmatrix}$$

$$\left(\begin{array}{l} C_1 \rightarrow C_1 - C_3 \\ C_2 \rightarrow C_2 - C_3 \end{array} \right)$$

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ c-a-b & c-a-b & (a+b)^2 \end{vmatrix}$$

$$= (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ 0 & c+a-b & b^2 \\ -2b & -2a & 2ab \end{vmatrix}$$

$$R_3 \rightarrow R_3 - (R_1 + R_2)$$

$$[= 2ab(c+a)(b+c-a) + 2a^2b(c+a-b)$$

$$= 2ab(c+a)(b+c-a) + (c+a-b)$$

$$= 2ab[(b(c+a) + (c^2-a^2) - ac + a^2 - ab)$$

$$= 2ab(bc + c^2 + ac)$$

$$= 2abc(a+b+c)$$

$$= 2abc(a+b+c)^3$$

60. The given determinant is

$$\begin{vmatrix} (a+1)(a+2) & (a+2) & 1 \\ (a+2)(a+3) & (a+3) & 1 \\ (a+3)(a+4) & (a+4) & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a^2+3a+2 & (a+2) & 1 \\ a^2+5a+6 & (a+3) & 1 \\ a^2+7a+12 & (a+4) & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a^2+3a+2 & (a+2) & 1 \\ 2a+4 & 1 & 0 \\ 4a+10 & 2 & 0 \end{vmatrix}$$

$$\begin{pmatrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{pmatrix}$$

$$= \begin{vmatrix} 2a+4 & 1 \\ 4a+10 & 2 \end{vmatrix}$$

$$= (4a+8) - (4a+10)$$

$$= -2$$

61. The given determinant is

$$\begin{vmatrix} b+c & c+a & a+b \\ a+b & b+c & c+a \\ c+a & a+b & b+c \end{vmatrix}$$

$$= \begin{vmatrix} 2(a+b+c) & c+a & a+b \\ 2(a+b+c) & b+c & c+a \\ 2(a+b+c) & a+b & b+c \end{vmatrix}$$

$$(C_1 \rightarrow C_1 + C_2 + C_3)$$

$$= 2 \begin{vmatrix} (a+b+c) & c+a & a+b \\ (a+b+c) & b+c & c+a \\ (a+b+c) & a+b & b+c \end{vmatrix}$$

$$= 2 \begin{vmatrix} (a+b+c) & -b & -c \\ (a+b+c) & -a & -b \\ (a+b+c) & -c & -a \end{vmatrix} \begin{pmatrix} C_2 \rightarrow C_2 - C_1 \\ C_3 \rightarrow C_3 - C_1 \end{pmatrix}$$

$$= 2 \begin{vmatrix} a & -b & -c \\ c & -a & -b \\ b & -c & -a \end{vmatrix} \quad (C_1 \rightarrow C_1 + C_2 + C_3)$$

$$= 2 \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

62 Here, $D = \begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix} = -4 - 9 = -13$

$$D_1 = \begin{vmatrix} 4 & 3 \\ 5 & -2 \end{vmatrix} = -8 - 15 = -23$$

$$D_2 = \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} = 10 - 12 = -2$$

$$\text{Thus, } x = \frac{D_1}{D} = \frac{-23}{-13} = \frac{23}{13}$$

$$\text{and } y = \frac{D_2}{D} = \frac{-2}{-13} = \frac{2}{13}$$

$$D = \begin{vmatrix} 1 & 3 \\ 2 & 6 \end{vmatrix} = 6 - 6 = 0$$

63. Here, $D_1 = \begin{vmatrix} 4 & 3 \\ 10 & 6 \end{vmatrix} = 24 - 30 = -6 \neq 0$

We know that, if $D = 0$ and any one of D_1 and D_2 is non-zero, it has no solution.

So the system of equation is inconsistent.

64. Here, $D = \begin{vmatrix} 2 & 5 \\ 6 & 15 \end{vmatrix} = 30 - 30 = 0$

$$D_1 = \begin{vmatrix} 6 & 5 \\ 18 & 15 \end{vmatrix} = 90 - 90 = 0$$

$$D_2 = \begin{vmatrix} 2 & 6 \\ 6 & 18 \end{vmatrix} = 36 - 36 = 0$$

As we know that, if $D = 0$ + $D_1 = D_2$, the system of equations has infinitely many solutions.

Let $y = k$

$$\text{Then } x = \frac{6-5k}{2}, k \in R.$$

65. Since the system of equations has infinitely many solutions, so $D = 0 + D_1 = D_2$

$$\text{Thus, } \begin{vmatrix} a & -b \\ (c+1) & c \end{vmatrix} = 0, \begin{vmatrix} 2a-b & -b \\ 10-a+3b & c \end{vmatrix} = 0$$

$$\text{and } \begin{vmatrix} a & 2a-b \\ c+1 & 10-a+3b \end{vmatrix} = 0$$

66. We have,

$$D = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$$

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ d & b & c \\ d^2 & b^2 & c^2 \end{vmatrix} = (d-b)(b-c)(c-d)$$

$$D_2 = \begin{vmatrix} 1 & 1 & 1 \\ a & d & c \\ a^2 & d^2 & c^2 \end{vmatrix} = (a-d)(d-c)(c-a)$$

$$D_3 = \begin{vmatrix} 1 & 1 & 1 \\ a & b & d \\ a^2 & b^2 & d^2 \end{vmatrix} = (a-b)(b-d)(d-a)$$

Thus,

$$x = \frac{D_1}{D} = \frac{(d-b)(b-c)(c-a)}{(a-b)(b-c)(c-a)} = \frac{(d-b)(c-a)}{(a-b)(c-a)}$$

$$y = \frac{D_2}{D} = \frac{(a-d)(d-c)(c-a)}{(a-b)(b-c)(c-a)} = \frac{(a-d)(d-c)}{(a-b)(b-c)}$$

$$z = \frac{D_3}{D} = \frac{(a-b)(b-d)(d-a)}{(a-b)(b-c)(c-a)} = \frac{(b-d)(d-a)}{(b-c)(c-a)}$$

67. Let $u = \frac{1}{x}, v = \frac{1}{y}, w = \frac{1}{z}$

The given system of equations reduces to

$$u + v + w = \frac{1}{4},$$

$$2u + v + 3w = \frac{9}{4}$$

and $-u - 2v + 4w = 1$

Here, $D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & 4 \end{vmatrix}$

$$= 1(-4+6) - 1(8+3) - 1(-4-1)$$

$$= 2 - 11 + 5 = -4$$

$$D_1 = \begin{vmatrix} 1/4 & 1 & 1 \\ 9/4 & 1 & 3 \\ 1 & 2 & 4 \end{vmatrix}$$

$$= \frac{1}{4} \cdot 2 \cdot (9-3) - \left(\frac{9}{2} + 1 \right)$$

$$= \frac{1}{2} + \frac{7}{2} = 4$$

Also, $D_2 = \begin{vmatrix} 1 & 1/4 & 1 \\ 2 & 9/4 & 3 \\ 1 & 1 & 4 \end{vmatrix}$

$$= (9-3) \cdot \frac{11}{4} - \left(2 + \frac{9}{4} \right)$$

$$= 4 - 5 = -1$$

Again, $D_3 = \begin{vmatrix} 1 & 1 & 1/4 \\ 2 & 1 & 9/4 \\ 1 & 2 & 1 \end{vmatrix}$

$$= \left(1 + \frac{9}{2} \right) - 1 \left(2 + \frac{9}{4} \right) = \frac{5}{4}$$

$$= \frac{7}{2} - 2 - \frac{14}{4} = 2$$

Now, $u = \frac{D_1}{D} = \frac{4}{-4} = -1 \Rightarrow x = \frac{1}{-1} = -1$

$$v = \frac{D_2}{D} = \frac{-1}{-4} = \frac{1}{4} \Rightarrow y = 4$$

and $w = \frac{D_3}{D} = \frac{2}{-4} = -\frac{1}{2} \Rightarrow z = -2$

68. Given parabola is $y = ax^2 + bx + c$, which is passing through (2, 4), (1, 1) and (2, 5), so,

$$4a + 2b + c = 4$$

$$a - b + c = 1$$

$$4a - 2b + c = 5$$

From Cramers rule,

$$a = \frac{D_1}{D} = \frac{15}{12} = \frac{5}{4}$$

$$b = \frac{D_2}{D} = \frac{1}{12}$$

$$c = \frac{D_3}{D} = \frac{2}{12} = \frac{1}{6}$$

where $D = \begin{vmatrix} 4 & 2 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{vmatrix} = 12$

$$D_1 = \begin{vmatrix} 4 & 2 & 1 \\ 1 & 1 & 1 \\ 5 & 2 & 1 \end{vmatrix} = 15$$

$$D_2 = \begin{vmatrix} 4 & 4 & 1 \\ 1 & 1 & 1 \\ 4 & 5 & 1 \end{vmatrix} = 1$$

$$D_3 = \begin{vmatrix} 4 & 2 & 4 \\ 1 & 1 & 1 \\ 4 & 2 & 5 \end{vmatrix} = 2$$

Hence, the required equation of the parabola is

$$y = ax^2 + bx + c$$

$$\Rightarrow y = \frac{5}{4}x^2 + \frac{x}{12} + \frac{1}{6}$$

69. Since the system of equations has a unique solution, so,

$$\frac{2}{3} \neq \frac{k}{-4}$$

$$\Rightarrow k \neq -\frac{8}{3}$$

Therefore, the value of k is $k \in \mathbb{R} - \left\{ -\frac{8}{3} \right\}$.

70. Since the system of equations has infinitely many solutions, so

$$\frac{3}{\lambda} = \frac{4}{8} = \frac{5}{10}$$

$$\Rightarrow \frac{3}{\lambda} = \frac{1}{2}$$

$$\Rightarrow \lambda = 6$$

Hence, the value of λ is 6.

71. We know that, the system of equations has no solution Only when, if $D = 0$, but any one of D_1, D_2, D_3 is non-zero.

Now, $D = 0$ gives

$$\begin{vmatrix} 1 & 2 & -3 \\ 0 & 0 & (p+2) \\ 0 & (2p+1) & 1 \end{vmatrix} = 0$$

$$\Rightarrow (p+2)(2p+1) = 0$$

$$\Rightarrow p = 2, -\frac{1}{2}$$

Now, $D_1 \neq 0$ gives $p \neq 1, -1$

$$D_2 \neq 0 \text{ gives } p \neq -\frac{1}{2}$$

$$\text{and } D_3 \neq 0 \text{ gives } p \neq -\frac{1}{2}$$

Hence, the value of p is -2 .

72. Since the given system of equations has no solution, so $D = 0$ and at-least any one of D_1, D_2, D_3 is non-zero.

Now $D = 0$, gives

$$D = \begin{vmatrix} 2 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 2 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 2(\lambda - 2) + (\lambda - 1) + 2(2 + 2) = 0$$

$$\Rightarrow -3\lambda - 4 - 1 + 8 = 0$$

$$\Rightarrow 3\lambda = 3$$

$$\Rightarrow \lambda = 1$$

Also $D_1 \neq 0$ gives

$$\begin{vmatrix} 2 & 1 & 2 \\ 4 & 2 & 1 \\ 4 & 2 & \lambda \end{vmatrix} \neq 0$$

$$\Rightarrow 2(-2\lambda - 2) + 2(-4\lambda - 4) \neq 0$$

$$\Rightarrow 4\lambda - 4 - 8\lambda - 8 \neq 0$$

$$\Rightarrow \lambda + 1 \neq 0$$

$$\Rightarrow \lambda \neq -1$$

Again, $D_2 \neq 0$ gives $\lambda \neq 1$.

Hence, the value of λ is $\emptyset$.

73. Here, $D = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 12 - 12 = 0$

So the system of equations has infinitely many solutions.

Let $y = k$

$$\text{Then } x = \frac{3k}{2}, k \in \mathbb{R}$$

74. Since the system of equations has non-trivial solution, so $D = 0$

$$\begin{vmatrix} a & t & b & c \\ b & c & t & a \\ c & a & a & t \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a+b+c & t & b & c \\ a+b+c-t & c-t & a & \\ a+b+c & t & a & a+t \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c) \begin{vmatrix} 1 & b & c \\ 1 & c & t \\ 1 & a & a+t \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c) \begin{vmatrix} 1 & b & c \\ 0 & c-b & t \\ 0 & a-b & a-c+t \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c) \begin{vmatrix} c & b-t & a & c \\ a & b & a & c-t \end{vmatrix} = 0$$

$$\Rightarrow (a+b+c) \begin{vmatrix} c & b & t & a & c \\ a & b & a & c & t \end{vmatrix} = 0$$

$$\Rightarrow t = (a+b+c), t^2 + (a+b)t + (2ac - c^2) = 0$$

$$\Rightarrow t = (a+b+c)$$

$$t = \frac{(a+b) \pm \sqrt{(a+b)^2 - 4(2ac - c^2)}}{2}$$

Thus, the number of values of t is 3.

75. Since the system of equations has a unique solution, so $D_1 \neq 0$

$$\begin{vmatrix} 6 & 5 & \lambda \\ 3 & -1 & 4 \\ 1 & 2 & 3 \end{vmatrix} \neq 0$$

$$\Rightarrow$$

$$\Rightarrow 6(3-8) - 5(-9-4) + \lambda(6+1) \neq 0$$

$$\Rightarrow -30 + 65 + 7\lambda \neq 0$$

$$\Rightarrow 7\lambda + 35 \neq 0$$

$$\Rightarrow \lambda + 5 \neq 0$$

$$\Rightarrow \lambda \neq -5$$

Hence, the value of λ is $\mathbb{R} - \{-5\}$.

76. Since the system of equations has infinite solutions, so

$$\begin{vmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 1 \end{vmatrix} = 0$$

$$\Rightarrow 1 - a(0 - a^2) = 0$$

$$\Rightarrow 1 - a^3 = 0$$

$$\Rightarrow (a+1)(a^2 - a + 1) = 0$$

$$\Rightarrow (a+1) = 0, (a^2 - a + 1) = 0$$

$$\Rightarrow a = -1, a = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm i\sqrt{3}}{2}$$

Hence, the values of a are $\left\{-1, \frac{1 \pm i\sqrt{3}}{2}\right\}$.

77. We have,

$$\begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix}^2 = \begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix} \times \begin{vmatrix} a & 0 & c \\ a & b & 0 \\ 0 & b & c \end{vmatrix}$$

$$= \begin{vmatrix} a.a+0.0+c.c & a.a+0.0+0.0 & a.0+0.b+c.c \\ a.a+b.0+0.c & a.a+b.b+0.0 & a.0+b.b+0.c \\ 0.a+b.0+c.c & 0.a+b.b+c.0 & 0.0+b.b+c.c \end{vmatrix}$$

$$= \begin{vmatrix} c^2+a^2 & a^2 & c^2 \\ a^2 & a^2+b^2 & b^2 \\ c^2 & b^2 & b^2+c^2 \end{vmatrix}$$

78. We have,

$$\begin{vmatrix} 2 & \alpha+\beta+\gamma+\delta & \alpha\beta+\gamma\delta \\ \alpha+\beta+\gamma+\delta & 2(\alpha+\beta)(\gamma+\delta) & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) & 2\alpha\beta\gamma\delta \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 0 \\ \alpha+\beta & \gamma+\delta & 0 \\ \alpha\beta & \gamma\delta & 0 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 0 \\ \gamma+\delta & \alpha+\beta & 0 \\ \gamma\delta & \alpha\beta & 0 \end{vmatrix}$$

$$= 0 \times 0$$

$$= 0$$

79. We have

$$\begin{vmatrix} 1 & \cos(\beta-\alpha) & \cos(\gamma-\alpha) \\ \cos(\alpha-\beta) & 1 & \cos(\gamma-\beta) \\ \cos(\alpha-\gamma) & \cos(\beta-\gamma) & 1 \end{vmatrix}$$

$$= \begin{vmatrix} \cos \alpha & \sin \alpha & 0 \\ \cos \beta & \sin \beta & 0 \\ \cos \gamma & \sin \gamma & 0 \end{vmatrix} \times \begin{vmatrix} \cos \alpha & \sin \alpha & 0 \\ \cos \beta & \sin \beta & 0 \\ \cos \gamma & \sin \gamma & 0 \end{vmatrix}$$

$$= 0 \times 0$$

$$= 0$$

80. We have,

$$\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix}$$

$$= \begin{vmatrix} 3 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}^2$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 0 & \alpha-1 & \beta-1 \\ 0 & \alpha^2-1 & \beta^2-1 \end{vmatrix}^2$$

$$= \begin{vmatrix} \alpha & 1 & \beta \\ \alpha^2-1 & \beta^2-1 & 1 \end{vmatrix}^2$$

$$= (\alpha-1)^2(\beta-1)^2 \begin{vmatrix} 1 & 1 \\ \alpha+1 & \beta+1 \end{vmatrix}^2$$

$$= (\alpha-1)^2(\beta-1)^2(\beta-\alpha)^2$$

$$= (1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$$

Hence, the value of k is 1.

81. We have,

$$F(x) = \begin{vmatrix} 0 & 0 & 0 \\ x & x^2 & x^3 \\ e^{x-a} & e^{x^2-a^2} & e^{x^3-a^3} \end{vmatrix}$$

$$+ \begin{vmatrix} 1 & a & a^2 \\ 1 & 2x & 3x^2 \\ e^{x-a} & e^{x^2-a^2} & e^{x^3-a^3} \end{vmatrix}$$

$$+ \begin{vmatrix} 1 & a & a^2 \\ x & x^2 & x^3 \\ e^{x-a} & 2x \times e^{x^2-a^2} & 3x^2 \times e^{x^3-a^3} \end{vmatrix}$$

$$\Rightarrow F'(a) = \begin{vmatrix} 1 & a & a^2 \\ 1 & 2a & 3a^2 \\ 1 & 1 & 1 \end{vmatrix} + \begin{vmatrix} 1 & a & a^2 \\ a & a^2 & a^3 \\ 1 & 2a & 3a^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a^2 \\ 1 & 2a & 3a^2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a^2 \\ 1 & 2a & 3a^2 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= 2a - 3a^2 - a + 3a^3 + a^2 - 2a^3$$

$$= a - 2a^2 + a^3$$

82. Given,

$$f(x) = \begin{vmatrix} 3 & 2 & 1 \\ 6x^2 & 2x^3 & x^4 \\ 1 & b & b^2 \end{vmatrix}$$

$$\Rightarrow f'(x) = \begin{vmatrix} 3 & 2 & 1 \\ 12x & 6x^2 & 4x^3 \\ 1 & b & b^2 \end{vmatrix}$$

$$\Rightarrow f'''(x) = \begin{vmatrix} 3 & 2 & 1 \\ 12 & 12x & 12x^2 \\ 1 & b & b^2 \end{vmatrix}$$

$$\Rightarrow f'''(b) = \begin{vmatrix} 3 & 2 & 1 \\ 12 & 12b & 12b^2 \\ 1 & b & b^2 \end{vmatrix}$$

$$\Rightarrow f'''(b) = 12 \begin{vmatrix} 3 & 2 & 1 \\ 1 & b & b^2 \\ 1 & b & b^2 \end{vmatrix} = 0$$

83. $\int_0^{\pi/2} f(x) dx$

$$= \left| \begin{array}{ccc} \int_0^{\pi/2} \sin^2 x dx & \int_0^{\pi/2} \log(\sin x) dx & \int_0^{\pi/2} \left(\frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} \right) dx \\ n & \sum_{k=1}^n (k) & \prod_{k=1}^n (k) \\ \frac{8}{15} & \frac{\pi}{2} \log\left(\frac{1}{2}\right) & \frac{\pi}{4} \end{array} \right|$$

$$= \left| \begin{array}{ccc} \frac{8}{15} & \frac{\pi}{2} \log\left(\frac{1}{2}\right) & \frac{\pi}{4} \\ n & \sum_{k=1}^n (k) & \prod_{k=1}^n (k) \\ \frac{8}{15} & \frac{\pi}{2} \log\left(\frac{1}{2}\right) & \frac{\pi}{4} \end{array} \right|$$

= 0

84. We have,

$$\Delta_r = \begin{vmatrix} r & 2012 & \frac{n(n+1)}{2} \\ 2r-1 & 2013 & n^2 \\ 3r-2 & 2014 & \frac{n(3n-1)}{2} \end{vmatrix}$$

$$\Rightarrow \sum_{r=1}^n \Delta_r = \begin{vmatrix} \sum_{r=1}^n r & 2012 & \frac{n(n+1)}{2} \\ \sum_{r=1}^n (2r-1) & 2013 & n^2 \\ \sum_{r=1}^n (3r-2) & 2014 & \frac{n(3n-1)}{2} \end{vmatrix}$$

$$= \begin{vmatrix} \frac{n(n+1)}{2} & 2012 & \frac{n(n+1)}{2} \\ n^2 & 2013 & n^2 \\ \frac{n(3n-1)}{2} & 2014 & \frac{n(3n-1)}{2} \end{vmatrix}$$

= 0

85. We have $D_r = \begin{vmatrix} 2^{r-1} & 101 & (2^n-1) \\ 3^{r-1} & 102 & \left(\frac{3^n-1}{2}\right) \\ 5^{r-1} & 103 & \left(\frac{5^n-1}{4}\right) \end{vmatrix}$

Then $\sum_{r=1}^n D_r = \begin{vmatrix} \sum_{r=1}^n 2^{r-1} & 101 & (2^n-1) \\ \sum_{r=1}^n 3^{r-1} & 102 & \left(\frac{3^n-1}{2}\right) \\ \sum_{r=1}^n 5^{r-1} & 103 & \left(\frac{5^n-1}{4}\right) \end{vmatrix}$

$$= \begin{vmatrix} (2^n-1) & 101 & (2^n-1) \\ \left(\frac{3^n-1}{2}\right) & 102 & \left(\frac{3^n-1}{2}\right) \\ \left(\frac{5^n-1}{4}\right) & 103 & \left(\frac{5^n-1}{4}\right) \end{vmatrix}$$

= 0

Then $\sum_{r=1}^n \Delta_r = \begin{vmatrix} \sum_{r=1}^n f(r) & \sum_{r=1}^n g(r) & \sum_{r=1}^n h(r) \\ a & b & c \\ p & q & r \end{vmatrix}$

where a, c, p, q, r are constants.

86. We have,

$$\text{adj}(A') = (\text{adj } A)'$$

$$\Rightarrow \text{adj}(A') - (\text{adj } A)' = \mathbf{O}$$

87. We know that,

$$|\text{adj}(A)| = |A|^{n-1}$$

Replace A by A^3 and n by 3, we get

$$|\text{adj}(A^3)| = |A^3|^{3-1} = |A|^6$$

88. We know that,

$$\text{given } A \cdot \text{adj}(A) = |A| \cdot I_n$$

$$= \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}$$

$$\Rightarrow |A| \cdot I_2 = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}$$

$$= 10 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= 10 \cdot I_2$$

$$\Rightarrow |A| = 10$$

89. We know that,

$$|\text{adj}(A)| = |A|^{n-1}$$

$$\Rightarrow |\text{adj}(A)| = |A|^{3-1} = |A|^2 = 16$$

90. We know that,

$$\begin{aligned} & |[\text{adj}\{\text{adj}(A)\}]| = |A|^{(n-1)^2} \\ \Rightarrow & |A|^{(n-1)^2} = |A|^{16} \\ \Rightarrow & (n-1)^2 = 16 \\ \Rightarrow & (n-1) = 4 \\ \Rightarrow & n = 5 \end{aligned}$$

91. Given $\text{adj}(P) = \begin{pmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{pmatrix}$

$$\Rightarrow |\text{adj}(P)| = \begin{vmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{vmatrix} = 4$$

As we know that,

$$|\text{adj}(P)| = |P|^{3-1} = |P|^2$$

Thus, $|P|^2 = 4$

$$\Rightarrow |P| = 2, -2$$

92. Let $P = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, where $|P| = 2$

Now, $Q = \begin{pmatrix} 2^2 a_{11} & 2^3 a_{12} & 2^4 a_{13} \\ 2^3 a_{21} & 2^4 a_{22} & 2^5 a_{23} \\ 2^4 a_{31} & 2^5 a_{32} & 2^6 a_{33} \end{pmatrix}$

$$\Rightarrow |Q| = \begin{vmatrix} 2^2 a_{11} & 2^3 a_{12} & 2^4 a_{13} \\ 2^3 a_{21} & 2^4 a_{22} & 2^5 a_{23} \\ 2^4 a_{31} & 2^5 a_{32} & 2^6 a_{33} \end{vmatrix}$$

$$= 2^2 \cdot 2^3 \cdot 2^4 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ 2a_{21} & 2a_{22} & 2a_{23} \\ 2^2 a_{31} & 2^2 a_{32} & 2^2 a_{33} \end{vmatrix}$$

$$= 2^2 \cdot 2^3 \cdot 2^4 \cdot 2 \cdot 2^2 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= 2^{12} \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= 2^{12} \cdot 2 = 2^{13}$$

93. We know that,

$$|\text{adj}(A)| = |A|^{n-1}$$

$$\Rightarrow |\text{adj}(A)| = |A|^2 = 16$$

$$\Rightarrow 1(12-12) - \alpha(4-6) + 3(4-6) = 16$$

$$\Rightarrow 2\alpha - 6 = 16$$

$$\Rightarrow 2\alpha = 16 + 6 = 22$$

$$\Rightarrow \alpha = 1$$

94. Since the determinant of a skew-symmetric matrix is zero, so the inverse does not exist

95. We know that,

$$BB^{-1} = I$$

$$\Rightarrow |BB^{-1}| = |I|$$

$$\Rightarrow |B||B^{-1}| = |I|$$

$$\Rightarrow |B^{-1}| = \frac{1}{|B|}$$

Now,

$$\begin{aligned} |B^{-1}AB| &= |B^{-1}||A||B| \\ &= \frac{1}{|B|} |A||B| = |A| \end{aligned}$$

96. We have,

$$R = (P \cos \theta + Q \sin \theta)$$

$$= \begin{pmatrix} \cos \theta & 0 \\ 0 & \cos \theta \end{pmatrix} + \begin{pmatrix} 0 & \sin \theta \\ -\sin \theta & 0 \end{pmatrix}$$

$$= \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$$

$$\text{Now, } R^{-1} = \frac{\text{adj}(R)}{|R|} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

97. Given $B^{-1}AB = A^2$

Now,

$$\begin{aligned} B^3 AB^3 &= B^2(B^{-1}AB)B^2 \\ &= B^{-2}(A^2)B^2 \\ &= B^{-1}(B^{-1}(A^2)B)B \\ &= B^{-1}(A^4)B \\ &= (A^4)^2 = A^8 \end{aligned}$$

98. Given,

$$I + A + A^2 + A^3 + \dots + A^k = O$$

$$\Rightarrow A^{-1}(I + A + A^2 + A^3 + \dots + A^k) = O$$

$$\Rightarrow (A^{-1} + I + A + A^2 + \dots + A^{k-1}) = O$$

$$\Rightarrow A^{-1} + (-A^k) = O$$

$$\Rightarrow A^{-1} = A^k$$

99. Given,

$$A^2 - A + I = O$$

$$\Rightarrow A^{-1}(A^2 - A + I) = A^{-1}O = O$$

$$\Rightarrow (A^{-1}A^2 - A^{-1}A + A^{-1}I) = O$$

$$\Rightarrow (A - I + A^{-1}) = O$$

$$\Rightarrow A^{-1} = I - A$$

100. We know that,

$$|\text{adj}(A)| = |A|^{n-1}$$

$$= |A|^{3-1} = |A|^2$$

$$\Rightarrow |\text{adj}(A^{-1})| = |A^{-1}|^2$$

$$\begin{aligned} \Rightarrow |\text{adj}(A^{-1})| &= \left(\frac{1}{|A|}\right)^2 = \frac{1}{|A|^2} \\ &= \frac{1}{|A|^2} = \frac{1}{25} \end{aligned}$$

101. We have,

$$|A^{-1}\text{adj}(A)| = |A^{-1}||\text{adj}(A)|$$

$$\begin{aligned}
 &= \frac{1}{|A|} |\text{adj}(A)| \\
 &= \frac{1}{|A|} \times |A|^{n-1} \\
 &= \frac{1}{|A|} \times |A|^2 \\
 &= |A|.
 \end{aligned}$$

102. Given,

$$A = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = 10 - 12 = -2$$

Since $|A| \neq 0$, so its inverse exists.

Thus,

$$\begin{aligned}
 A^{-1} &= \frac{\text{adj}(A)}{|A|} \\
 &= -\frac{1}{2} \begin{pmatrix} 5 & -3 \\ -4 & 2 \end{pmatrix} \\
 &= \begin{pmatrix} -5/2 & 3/2 \\ 2 & -1 \end{pmatrix}
 \end{aligned}$$

103. Given $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \cdot X = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$

It can be written as $AX = B$, where

$$A = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1 \neq 0$$

So, its inverse exists.

Thus,

$$\begin{aligned}
 X &= A^{-1}B \\
 &= \begin{pmatrix} 1 & -3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \\
 &= \begin{pmatrix} 1 & 4 \\ 0 & -1 \end{pmatrix}.
 \end{aligned}$$

104. We have,

$$B^{-1} = \frac{\text{adj}(B)}{|B|} = \frac{1}{2} \begin{pmatrix} 3 & 1 \\ -7 & 3 \end{pmatrix}$$

$$\Rightarrow 2B^{-1} = \begin{pmatrix} 3 & -1 \\ 7 & 3 \end{pmatrix}$$

Now

$$\begin{aligned}
 \det(2A^9 B^{-1}) &= \det(A^9 2B^{-1}) \\
 &= \det(A^9) \times \det(2B^{-1}) \\
 &= |A|^9 \times |2B^{-1}| \\
 &= (-1)^9 \times (9 - 7) \\
 &= -2
 \end{aligned}$$

105. We know that,

$$B^{-1} = \frac{\text{adj}(B)}{|B|} = \frac{A}{|B|}$$

$$\Rightarrow \text{adj}(B) = |B|B^{-1}$$

Replacing B by QBP , we get

$$\text{adj}(QBP) = |QBP|(QBP)^{-1}$$

$$\Rightarrow \text{adj}(QBP) = |Q||B||P|P^{-1}B^{-1}Q^{-1}$$

$$\Rightarrow \text{adj}(QBP) = |B|P^{-1}B^{-1}Q^{-1}$$

$$\Rightarrow \text{adj}(QBP) = P^{-1}|B|B^{-1}Q^{-1}$$

$$\Rightarrow \text{adj}(QBP) = P^{-1}(A)Q^{-1}$$

106. We have,

$$B = A^{-1}A'$$

$$\Rightarrow AB = A'$$

Now,

$$ABB' = A'B'$$

$$\Rightarrow ABB' = (BA)' = (A^{-1}AA')'$$

$$\Rightarrow = (IA')' = (A')' = A$$

$$\Rightarrow BB' = 1$$

107. Now,

$$\begin{aligned}
 |A| &= \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 3 & 2 & 4 \end{vmatrix} \\
 &= 0 - 2(8 - 6) + 3(4 - 3) \\
 &= -4 - 3 \\
 &= -7 \neq 0
 \end{aligned}$$

So, its inverse exists.

We have,

$$\begin{aligned}
 \text{adj}(A) &= \begin{pmatrix} \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} & \begin{vmatrix} 2 & 2 \\ 3 & 4 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} \\ \begin{vmatrix} 2 & 3 \\ 2 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 3 & 2 \end{vmatrix} \\ \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 2 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \end{pmatrix}^T \\
 &= \begin{pmatrix} 0 & -2 & 1 \\ -2 & -5 & 4 \\ 1 & 4 & 3 \end{pmatrix}^T \\
 &= \begin{pmatrix} 0 & -2 & 1 \\ -2 & -5 & 4 \\ 1 & 4 & 3 \end{pmatrix}
 \end{aligned}$$

$$\text{Thus, } A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$= -\frac{1}{7} \begin{pmatrix} 0 & -2 & 1 \\ -2 & -5 & 4 \\ 1 & 4 & -3 \end{pmatrix}$$

108. We have,

$$(I + A) = \begin{pmatrix} 1 & -\tan\left(\frac{\alpha}{2}\right) \\ \tan\left(\frac{\alpha}{2}\right) & 1 \end{pmatrix}$$

and

$$(I - A) = \begin{pmatrix} 1 & \tan\left(\frac{\alpha}{2}\right) \\ -\tan\left(\frac{\alpha}{2}\right) & 1 \end{pmatrix}$$

Let $I - A = B$

$$\text{Now, } B^{-1} = \frac{\text{adj}(B)}{|B|}$$

$$= \frac{1}{\sec^2\left(\frac{\alpha}{2}\right)} \begin{pmatrix} 1 & -\tan\left(\frac{\alpha}{2}\right) \\ \tan\left(\frac{\alpha}{2}\right) & 1 \end{pmatrix}$$

Thus,

$$(I - A)^{-1} (I + A) = B^{-1} (I + A)$$

$$= \frac{1}{\sec^2\left(\frac{\alpha}{2}\right)} \begin{pmatrix} 1 & -\tan\left(\frac{\alpha}{2}\right) \\ \tan\left(\frac{\alpha}{2}\right) & 1 \end{pmatrix} \begin{pmatrix} 1 & -\tan\left(\frac{\alpha}{2}\right) \\ \tan\left(\frac{\alpha}{2}\right) & 1 \end{pmatrix}$$

$$= \frac{1}{\sec^2\left(\frac{\alpha}{2}\right)} \times \begin{pmatrix} 1 - \tan^2\left(\frac{\alpha}{2}\right) & -2 \tan\left(\frac{\alpha}{2}\right) \\ 2 \tan\left(\frac{\alpha}{2}\right) & 1 - \tan^2\left(\frac{\alpha}{2}\right) \end{pmatrix}$$

$$= \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$$

109. The given system of equations can be written in the matrix form as

$AX = B$, where

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \text{ and } B = \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$$

$$\text{Now, } |A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 6 \neq 0$$

Thus, A^{-1} exists

$$\text{Now, } A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$= \frac{1}{6} \begin{pmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{pmatrix}$$

Therefore, $X = A^{-1} B$

$$= \frac{1}{6} \begin{pmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{pmatrix} \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix}$$

$$= \frac{1}{6} \begin{pmatrix} 6 \\ 12 \\ 18 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\Rightarrow x = 1, y = 2 \text{ and } z = 3.$$

110. Given matrix is

$$A = \begin{pmatrix} -1 & 2 & 5 \\ 2 & -3 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} -1 & 2 & 5 \\ 2 & -3 & 1 \\ -1 & 1 & 1 \end{vmatrix} = 11 \neq 0$$

$\Rightarrow A^{-1}$ exists.

Thus,

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$= \frac{1}{11} \begin{pmatrix} 7 & 2 & -6 \\ -2 & 1 & -3 \\ -4 & 2 & 5 \end{pmatrix}$$

Also, the given system of equations can be written in matrix form as

$$\begin{pmatrix} -1 & 2 & 5 \\ 2 & -3 & 1 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 15 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 & 2 & 5 \\ 2 & -3 & 1 \\ -1 & 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 2 \\ 15 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{11} \begin{pmatrix} 7 & 2 & -6 \\ -2 & 1 & -3 \\ -4 & 2 & 5 \end{pmatrix} \begin{pmatrix} 2 \\ 15 \\ 3 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{1}{11} \begin{pmatrix} 44 \\ -33 \\ 11 \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 1 \end{pmatrix}$$

$$\Rightarrow x = 4, y = -3 \text{ and } z = 1.$$

111. The given system of equations can be written in matrix form as

$$\begin{pmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \\ 5 \end{pmatrix}$$

$\Rightarrow AX = B$, where

$$A = \begin{pmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{pmatrix}, X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, B = \begin{pmatrix} 4 \\ 9 \\ 5 \end{pmatrix}$$

Now, $|A| = \begin{vmatrix} 5 & 3 & 7 \\ 3 & 26 & 2 \\ 7 & 2 & 10 \end{vmatrix} = 0$

So, inverse does not exist.

Also,

$$\text{adj}(A) = \begin{pmatrix} 256 & -16 & -176 \\ -16 & 1 & 11 \\ -176 & 11 & 121 \end{pmatrix}$$

Thus,

$$\begin{aligned} \text{adj}(A)B &= \begin{pmatrix} 256 & -16 & -176 \\ -16 & 1 & 11 \\ -176 & 11 & 121 \end{pmatrix} \begin{pmatrix} 4 \\ 9 \\ 5 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \mathbf{O} \end{aligned}$$

Therefore, the given system of equations have infinitely many solutions.

112. The given system of equations can be written in matrix form as

$$AX = B$$

It has either

- (i) a unique solution, or
- (ii) infinite solutions, or
- (iii) no solution.

Thus, there cannot exist any matrix A such that

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \text{ has two distinct solutions.}$$

113. Let $\frac{x^2}{a^2} = X$, $\frac{y^2}{b^2} = Y$ and $\frac{z^2}{c^2} = Z$.

The given system of equations reduces to

$$X + Y - Z = 1$$

$$X - Y + Z = 1$$

$$X + Y + Z = 1$$

It can be written in matrix form as

$$\begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Let $AX' = B$, where

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, X' = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}, B = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Now,

$$|A| = \begin{vmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = -4 \neq 0$$

Thus, the system of equations have a unique solution.

114. We have,

$$A = IA$$

$$\Rightarrow \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot A$$

$$\Rightarrow \begin{pmatrix} 1 & 2 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} \cdot A$$

$$(R_2 \rightarrow R_2 - 2R_1)$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ -3 & 1 \end{pmatrix} \cdot A$$

$$(R_1 \rightarrow R_1 + 2R_2)$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix} \cdot A$$

$$\left[R_2 \rightarrow \left(\frac{-1}{2} \right) \times R_2 \right]$$

$$\text{Thus, } A^{-1} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}.$$

115. We have,

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{vmatrix}$$

$$\begin{aligned} &= 1(40 - 42) - 4(20 - 21) + 3(12 - 12) \\ &= -2 + 2 + 0 = 0 \end{aligned}$$

Since the determinant of A is zero, so the rank of the given matrix is 2.

116. We have,

$$|A| = \begin{vmatrix} 2 & 4 & 3 \\ 1 & 2 & -1 \\ -1 & -2 & 6 \end{vmatrix}$$

$$\begin{aligned} &= 2(12 - 2) - 4(6 - 1) + 3(-2 + 2) \\ &= 20 - 20 + 0 \\ &= 0 \end{aligned}$$

Since the determinant of A is zero, so the rank of the given matrix is 2.

117. We have,

$$|A| = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{vmatrix}$$

$$\begin{aligned} &= 1(18 - 20) - 2(12 - 12) + 3(10 - 12) \\ &= -2 - 6 = -8 \end{aligned}$$

Since the determinant of A is non-zero, so the rank of the given matrix is 3.

118. We have,

$$A = \begin{bmatrix} 1 & -2 & 1 & -1 \\ 1 & 1 & -2 & 3 \\ 4 & 1 & -5 & 6 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 1 & -2 & 1 & -1 \\ 0 & 3 & -3 & 4 \\ 0 & 9 & 9 & 12 \end{bmatrix}$$

$$(R_2 \rightarrow R_2 \quad R_1 \cdot R_3 \rightarrow R_3 \quad 4R_1)$$

$$\Leftrightarrow \begin{bmatrix} 1 & -2 & 1 & -1 \\ 0 & 1 & -1 & 4/3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(R_3 \rightarrow R_3 - 9R_2)$$

Since the number of non-zero rows is 2, so the rank of the given matrix is 2.

119. We have,

$$\begin{aligned} A^2 &= A \cdot A = \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} \\ &= \begin{bmatrix} 4+3 & 5 & 6 & 12+15 & 10 & 15+20 \\ -2-4+5 & 3+16-15 & 5+20 & 20 \\ 2+3 & 4 & 3 & 12+12 & 5 & 15+16 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -3 & -5 \\ -1 & 4 & 5 \\ 1 & -3 & -4 \end{bmatrix} = A \end{aligned}$$

Thus, A is idempotent.

120. We have,

$$A^2 = A \cdot A$$

$$\begin{aligned} &= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & 2 & 1 \end{bmatrix} = A. \end{aligned}$$

Hence, A is periodic.

121. Let $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ 2 & 1 & 3 \end{bmatrix}$.

$$\begin{aligned} \text{Now, } A^2 &= A \cdot A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & 1 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ 1 & 1 & -3 \end{bmatrix} \end{aligned}$$

Also, $A^3 = A^2 \cdot A$

$$\begin{aligned} &= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ 1 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ 2 & 1 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ &= O \end{aligned}$$

Hence A is nilpotent of order 3.

122. Let $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$

Now, $A^2 = A \cdot A$

$$\begin{aligned} &= \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Hence A is involutory.

123. Give, $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$

For orthogonal matrix $AA^T = A^T A = I$

$$\begin{aligned} \therefore AA^T &= \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \end{aligned}$$

Hence A is an orthogonal matrix.

124. We have,

$$A^T = \begin{bmatrix} 3 & 3+4i & 5 & 2i \\ 3 & 4i & 5 & 2i \\ 5+2i & -2+i & 2 & \end{bmatrix}$$

$$\begin{aligned} \Rightarrow \overline{(A^T)} &= \begin{bmatrix} 3 & 3 & 4i & 5+2i \\ 3+4i & 5 & 2+i \\ 5-2i & -2-i & 2 \end{bmatrix} \\ &= A \end{aligned}$$

Thus, A is a hermitian matrix.

125 We have,

$$\Rightarrow A^T = \begin{bmatrix} 2i & -2-3i & 2+i \\ -2-3i & -i & 3i \\ 2+i & 3i & 0 \end{bmatrix}$$

$$(\overline{A^T}) = \begin{bmatrix} -2i & 2+3i & 2-i \\ -2+3i & -i & -3i \\ -2-i & -3i & 0 \end{bmatrix}$$

$$\Rightarrow (\overline{A^T}) = -\begin{bmatrix} 2i & -2-3i & 2+i \\ -2-3i & -i & 3i \\ 2+i & 3i & 0 \end{bmatrix} = -A$$

Hence A is an skew Hermitian matrix.

126. Let $A = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$

$$\Rightarrow A^\theta = \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$\Rightarrow AA^\theta = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix} \times \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1+i \\ 1-i & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= I$$

Hence A is unitary matrix.